



Clinical Operations Manual

HOSPITAL MANAGEMENT SYSTEM V1.0

Technical Infrastructure

- **Architecture:** Fully decoupled React 19 / FastAPI bridge
- **Backbone:** Python High-Performance Async Core
- **Storage:** SQLite Relational Schema
- **Security:** X-User-Id Multi-factor Identity Context

****Frontend****

- **Framework:** React 19 (via Vite)
- **Styling:** Vanilla CSS (Custom Design System)
- **State Management:** React Hooks & Context API
- **Routing:** React Router v7

Backend API

- **Runtime:** Python 3.10+
- **Framework:** FastAPI
- **Authentication:** Custom `X-User-Id` Header-based Authentication
- **Database:** SQLite 3 (Persistent storage via `hms.db`)
- **Dependencies:** SQLAlchemy, Pydantic, Uvicorn

🗄 Database Schema

The system utilizes a relational SQLite database with the following core entities:

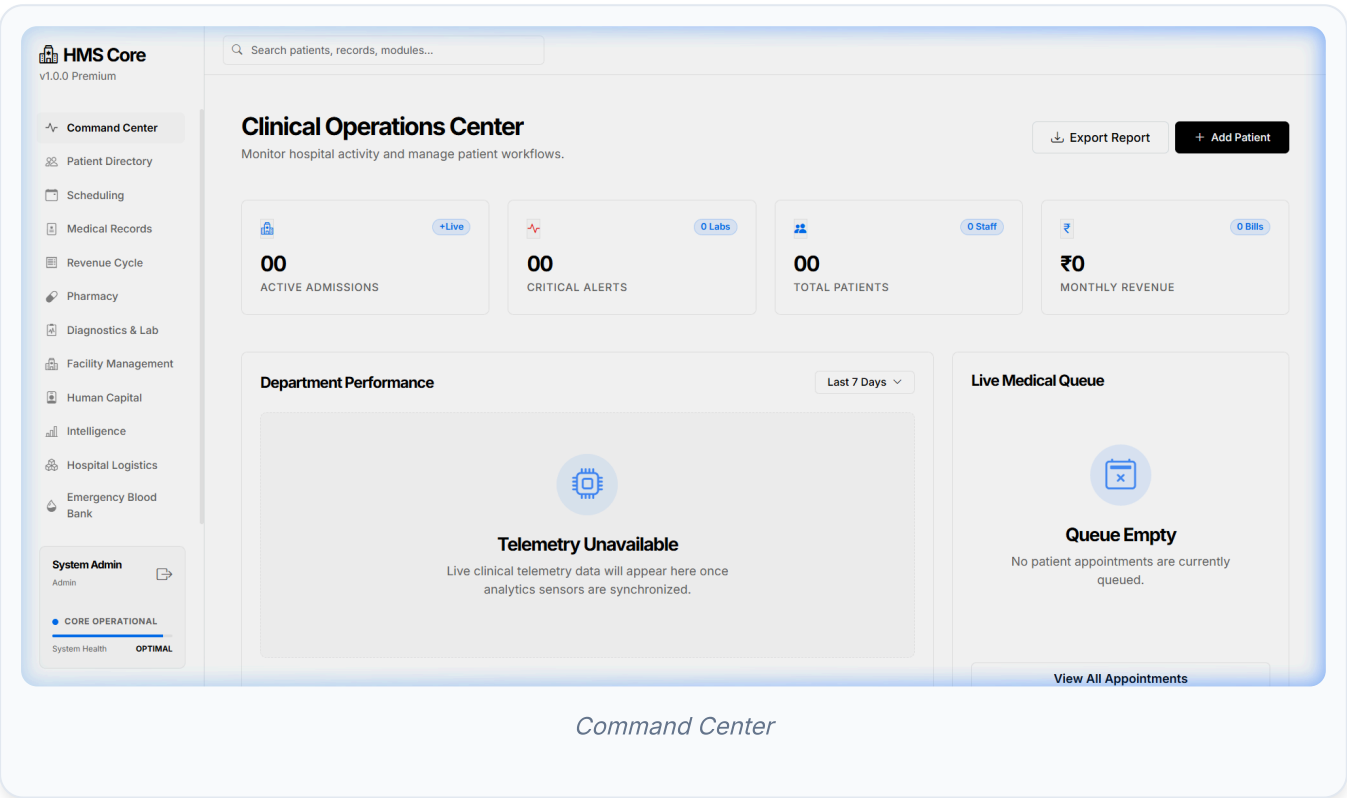
Table	Description	--- ---	`users`	System administrators and staff login credentials.
	`patients`	Comprehensive registry of patient profiles and medical status.		
	`appointments`	Scheduling records for patient-doctor consultations.		
	`records`	Clinical histories, diagnoses, and prescriptions.		
	`invoices`	Billing and revenue tracking data.		
	`medicines`	Pharmacy inventory and stock management.		
	`staff`	Human Capital management (Doctors, Nurses, Admin).		
	`blood_inventory`	Real-time tracking of blood groups and units.		
	`inventory`	General hospital logistics and supply tracking.		

🏥 Clinical & Operational Modules

The Hospital Management System comprises 13 specialized modules, each engineered for clinical precision and operational efficiency.

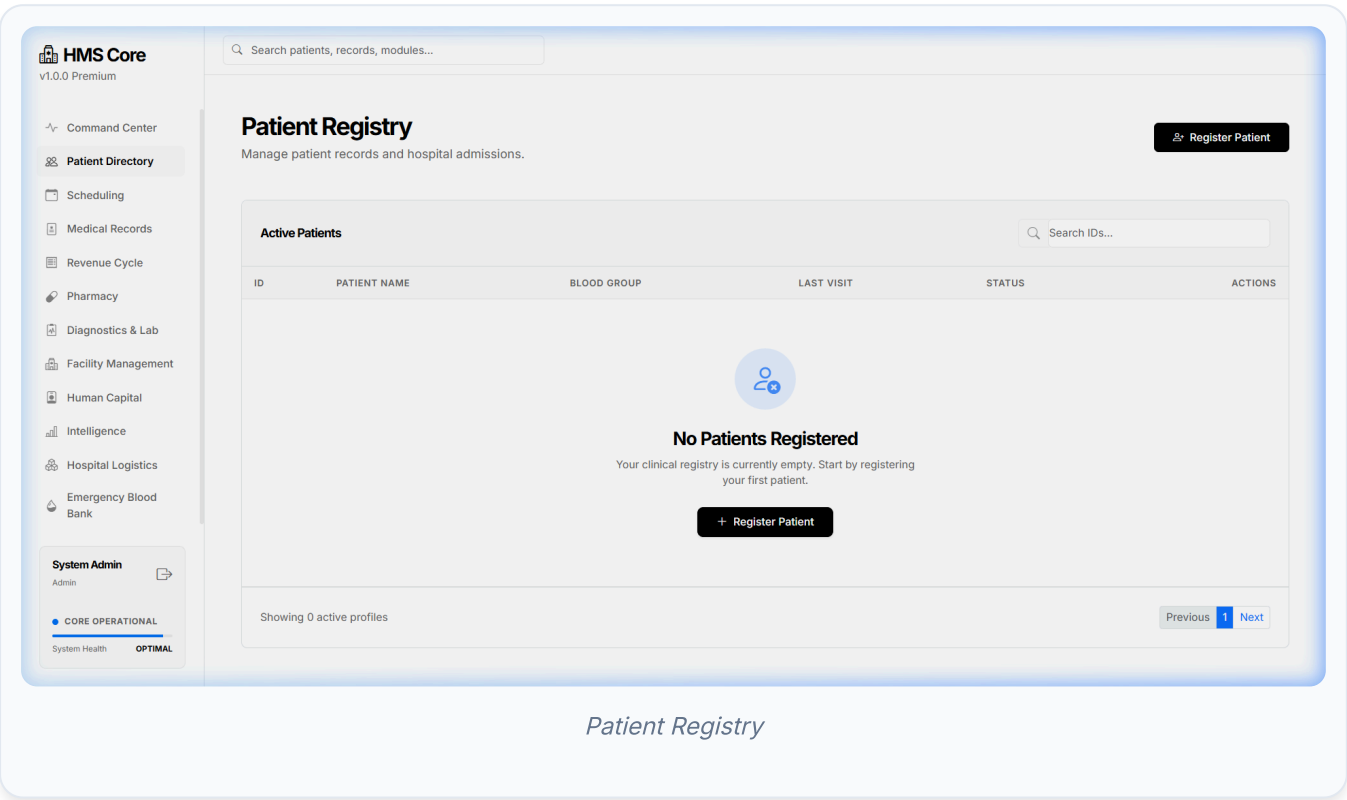
1. Command Center (Dashboard)

The central administrative hub provides real-time telemetry on hospital operations. It features dynamic KPIs for active admissions, critical patient alerts, laboratory throughput, and financial revenue streams.



2. Patient Registry (EMR)

The Electronic Medical Record (EMR) core manages the full patient lifecycle—from initial registration and triage to long-term clinical history tracking. It ensures data integrity and rapid access to patient demographics and blood group compatibility.



****3. Appointment Scheduling****

A streamlined scheduling engine for managing outpatient consultations. It prevents double-booking and optimizes doctor-patient interaction times across various clinical departments.

****4. Medical Records & Diagnoses****

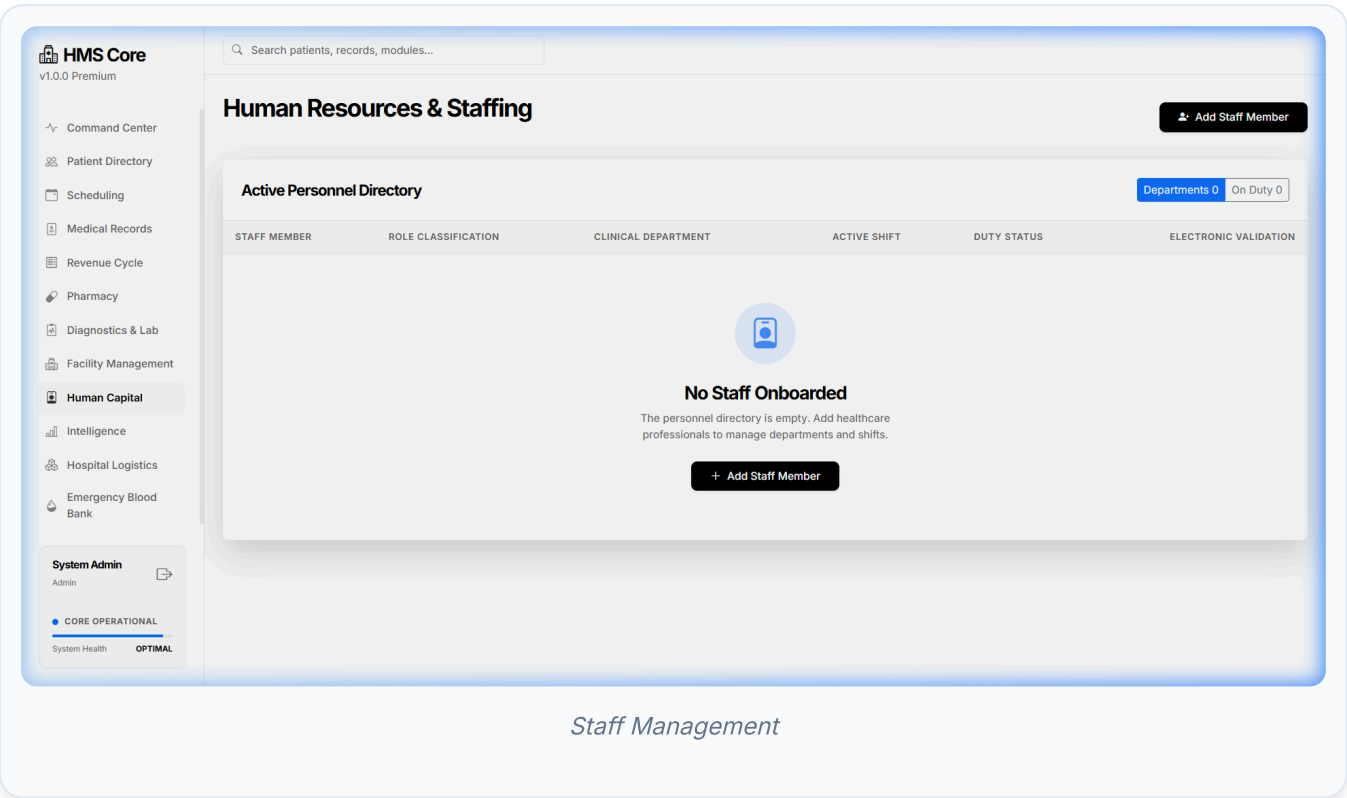
This module stores digitized clinical notes, formal diagnoses, and comprehensive prescription histories. It serves as the primary reference for physicians during follow-up consultations.

****5. Financial Operations (Invoices)****

The billing engine automates the generation of medical invoices and tracks revenue across different payment methods. It provides transparency in hospital accounting and patient billing cycles.

6. Human Capital (Staff)

A comprehensive directory for managing hospital personnel, including clinical staff (Doctors, Nurses) and administrative teams. It tracks departmental assignments, active shifts, and operational status.

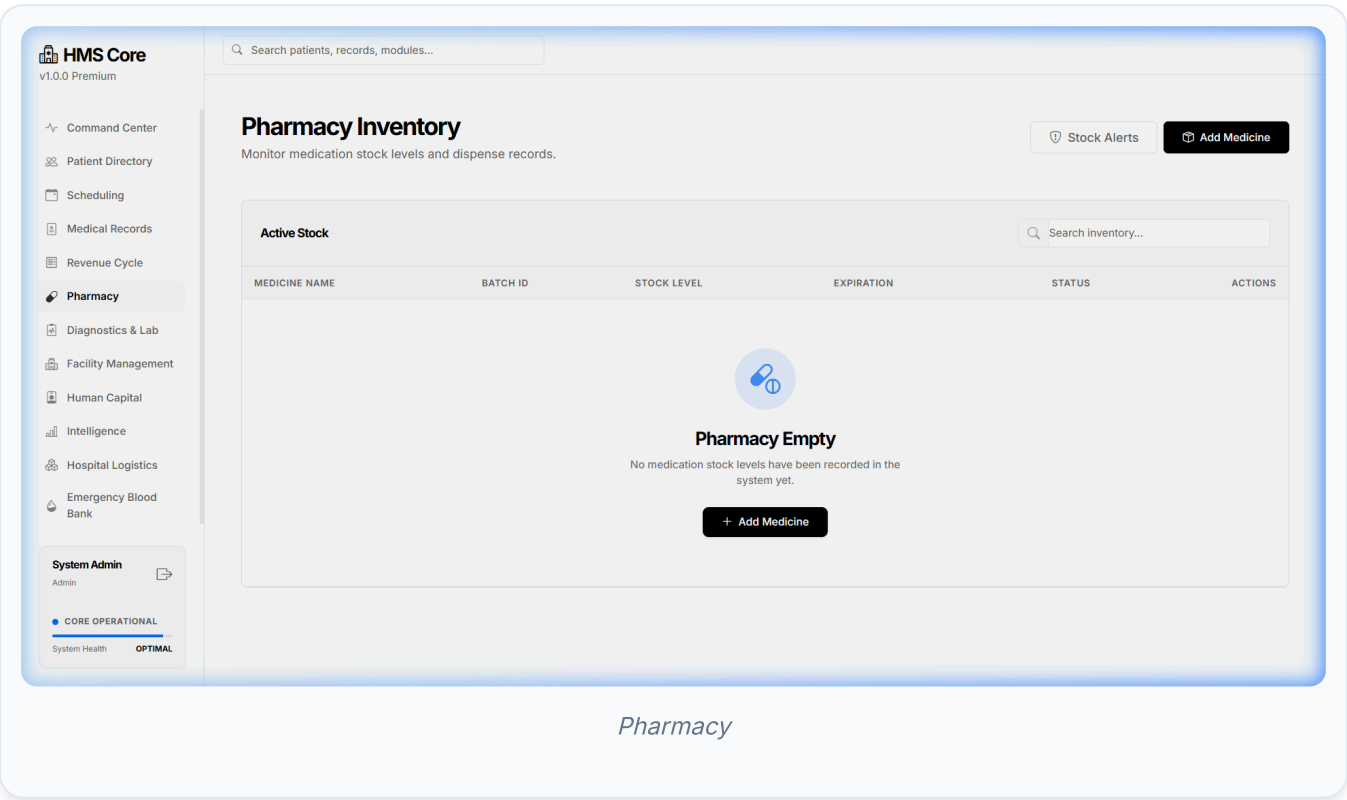


****7. Diagnostic Testing (Labs)****

Manages the lifecycle of laboratory orders and results. It tracks patient samples from collection to final diagnostic reporting, ensuring high throughput and accuracy in clinical testing.

8. Clinical Pharmacy

Automated pharmacy management for tracking medicinal stock, batch numbers, and expiry dates. It integrates with clinical prescriptions to ensure accurate medication dispensing.



9. Emergency Blood Bank

A critical lifecycle module for managing blood supply. It monitors units of all major blood groups (A+, B-, O+, etc.) in real-time and provides visual alerts for low-stock scenarios.

HMS Core

v1.0.0 Premium

Command Center

Patient Directory

Scheduling

Medical Records

Revenue Cycle

Pharmacy

Diagnostics & Lab

Facility Management

Human Capital

Intelligence

Hospital Logistics

Emergency Blood Bank

System Admin

Admin

CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Blood Bank Repository

Monitor real-time blood unit availability and donation cycle.

Request Registry

Add Donor

A+

0 Units

STABLE

Stock Trend: +0 today

B-

0 Units

STABLE

Stock Trend: +0 today

O+

0 Units

STABLE

Stock Trend: +0 today

B+

0 Units

STABLE

Stock Trend: +0 today

A-

0 Units

STABLE

Stock Trend: +0 today

AB-

0 Units

STABLE

Stock Trend: +0 today

AB+

0 Units

STABLE

Stock Trend: +0 today

O-

0 Units

STABLE

Stock Trend: +0 today

Recent Activity Logs

TYPE	GROUP	UNITS	ENTITY/PERSON	TIMESTAMP
[Empty table body]				

Blood Bank

****10. Inpatient Management (Admissions)****

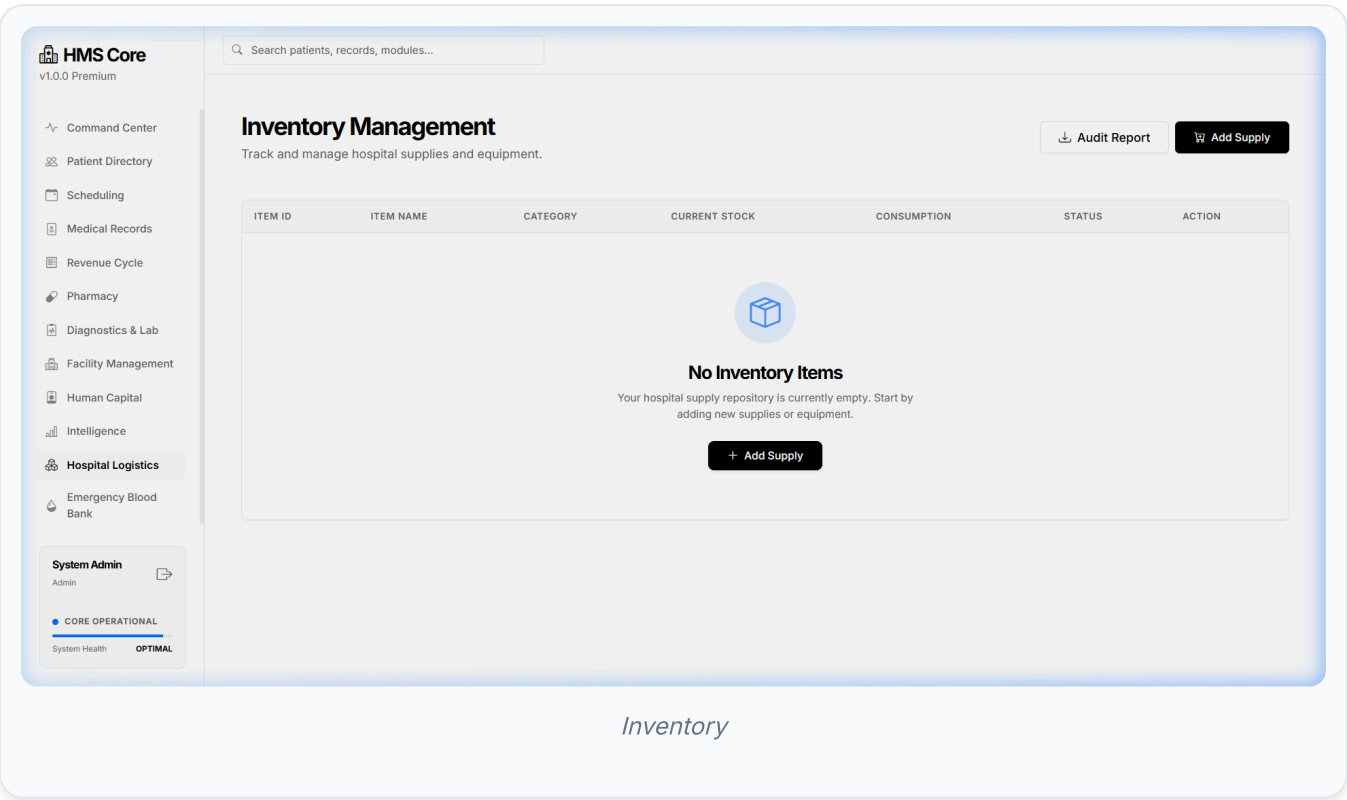
Handles the admission and discharge workflows for long-term patients. It integrates with ward management to optimize hospital bed utilization.

****11. Ward & Bed Management****

Provides a real-time map of hospital wards, bed availability, and occupancy rates. It ensures efficient patient placement based on clinical urgency and department capacity.

12. Hospital Logistics (Inventory)

A robust supply chain engine for managing non-medicinal inventory, from surgical equipment to general clinical supplies. It features automated alerts for stock replenishment.



****13. Operational Insights (Reports)****

The analytics engine generates a dashboard of operational health, summarizing departmental performance, resource allocation, and clinical efficiency metrics.

🚀 Execution & Deployment

****Frontend Development****

``bash npm run dev `` Starts the Vite development server on `http://localhost:5173`.

****Backend Development****

``bash cd backend python -m uvicorn app.main:app --reload --port 8000 `` Runs the FastAPI server on `http://localhost:8000`.

--- > [!NOTE] > This document was generated automatically by Antigravity based on a real-time audit of the codebase and UI.